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Welcome to the Lab n°5



0

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Today's Planning

- Menti : neural networks + text analytics
- Walkthrough text analytics pt.1
- Presentation assignment 4
- Q&A exam questions released last week

0

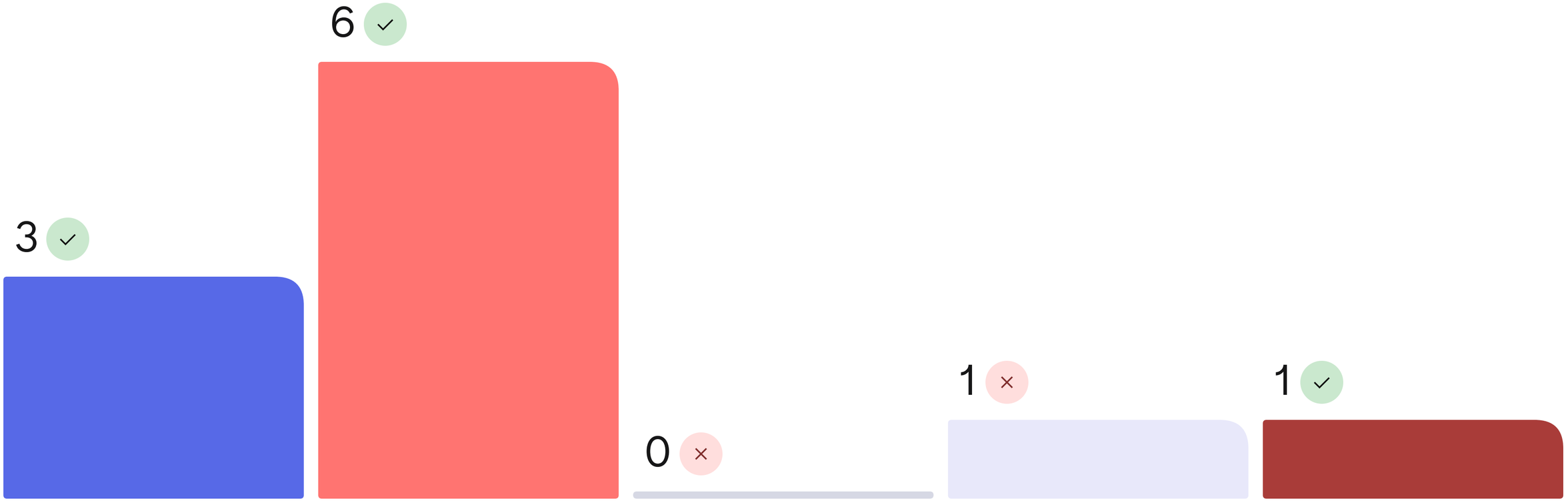
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Walkthrough 4

(Neural Networks)

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What is the idea behind neural networks



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Can you think of some uses for Neural Networks ?

image analysis

identification

classification

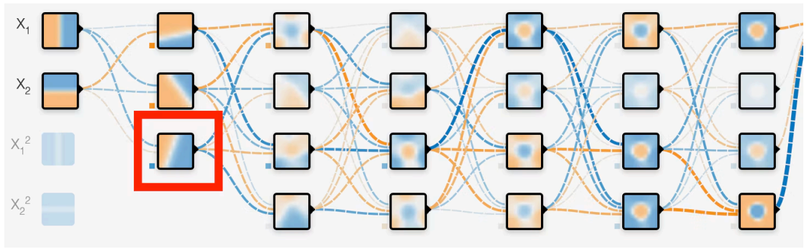
find if a cat on a pictur

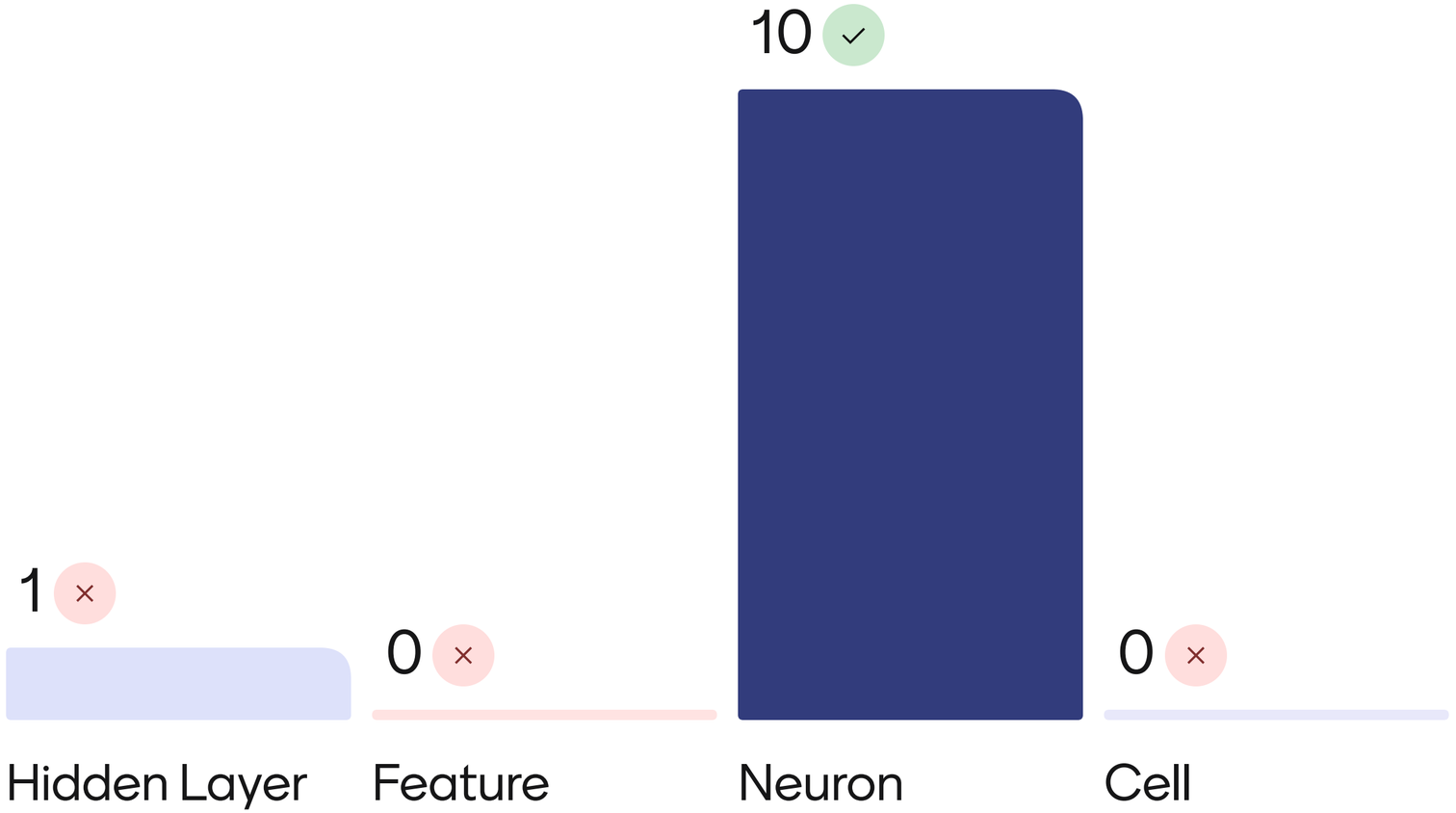
stock predictions

5/5

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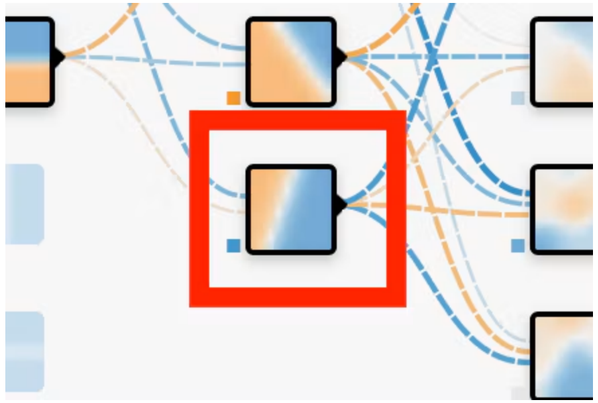
What is the name of the part highlighted below?





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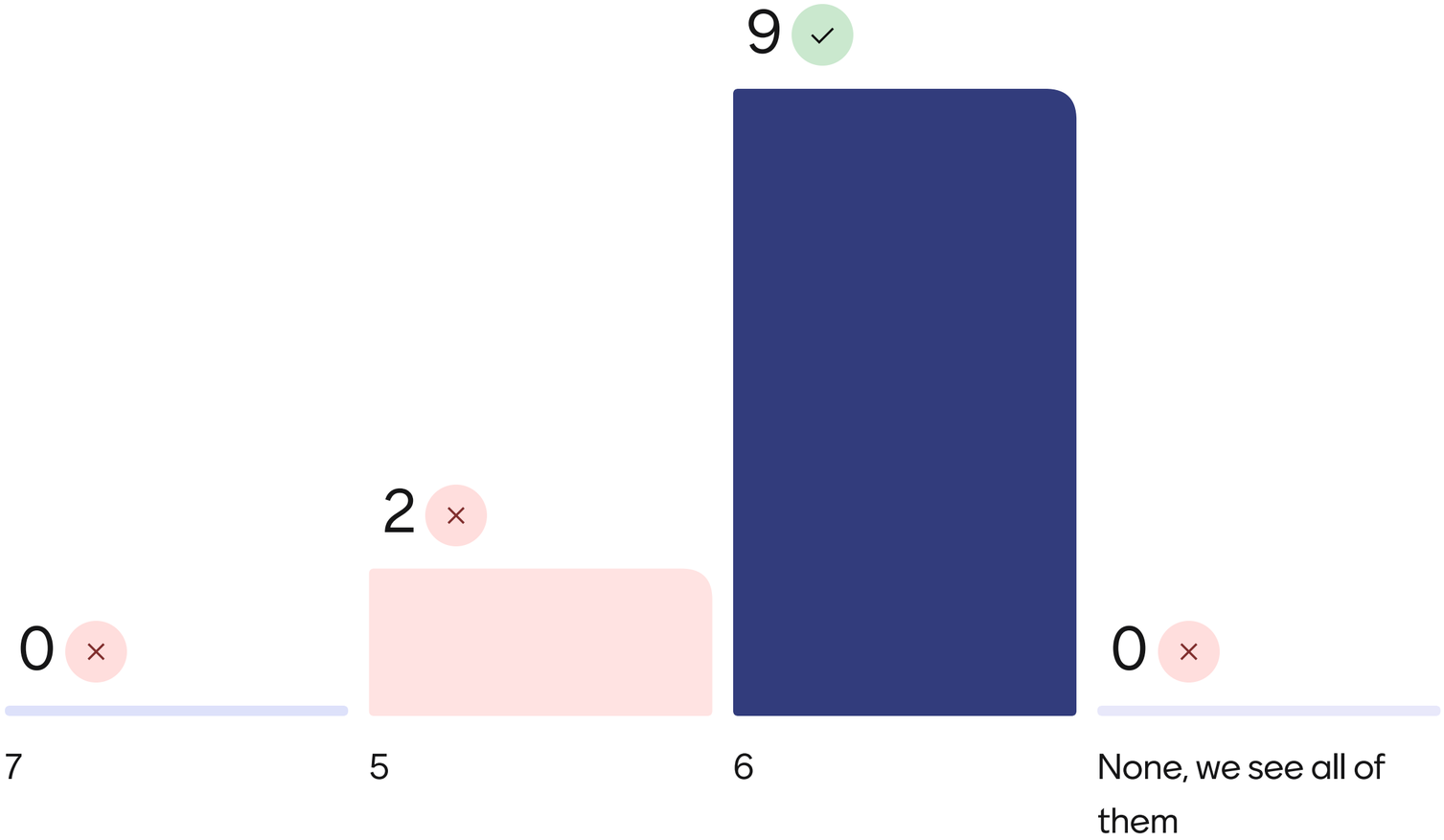
What does a neuron do in a neural network?

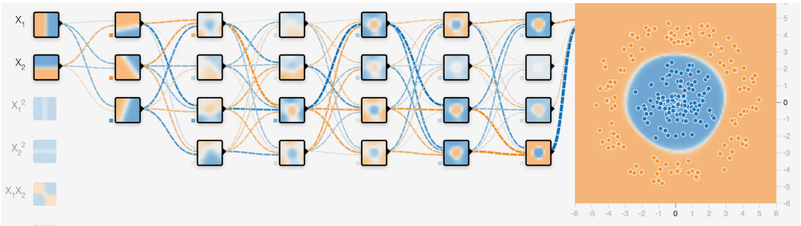


0	Memorizes examples from the dataset
10	A computational unit that receives inputs, applies weights, and outputs a value
0	Generates new data samples

10/10

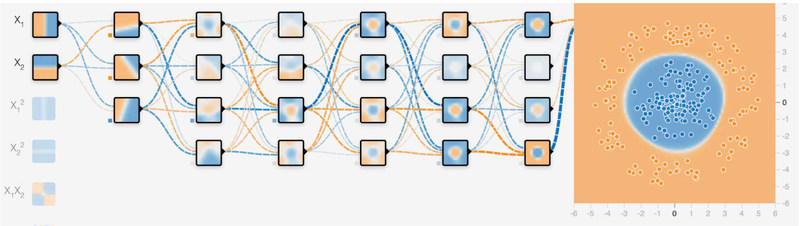
How many hidden layers are there?





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What is the main role of the hidden layers in a neural network?

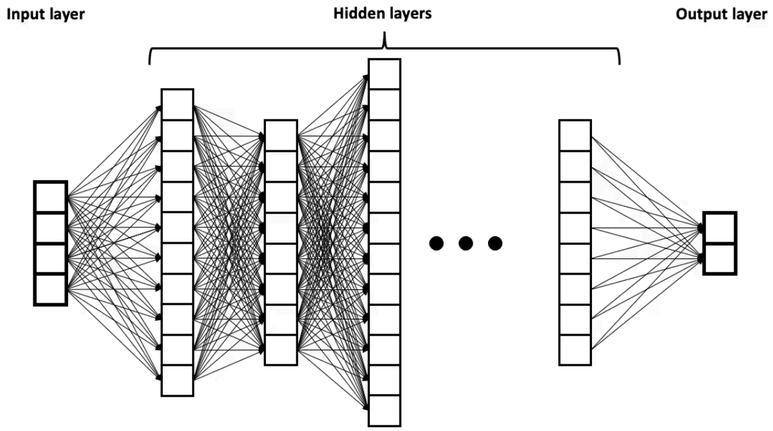


0	0	0	0
Input data storage	Learning features	Output prediction	Data normalization

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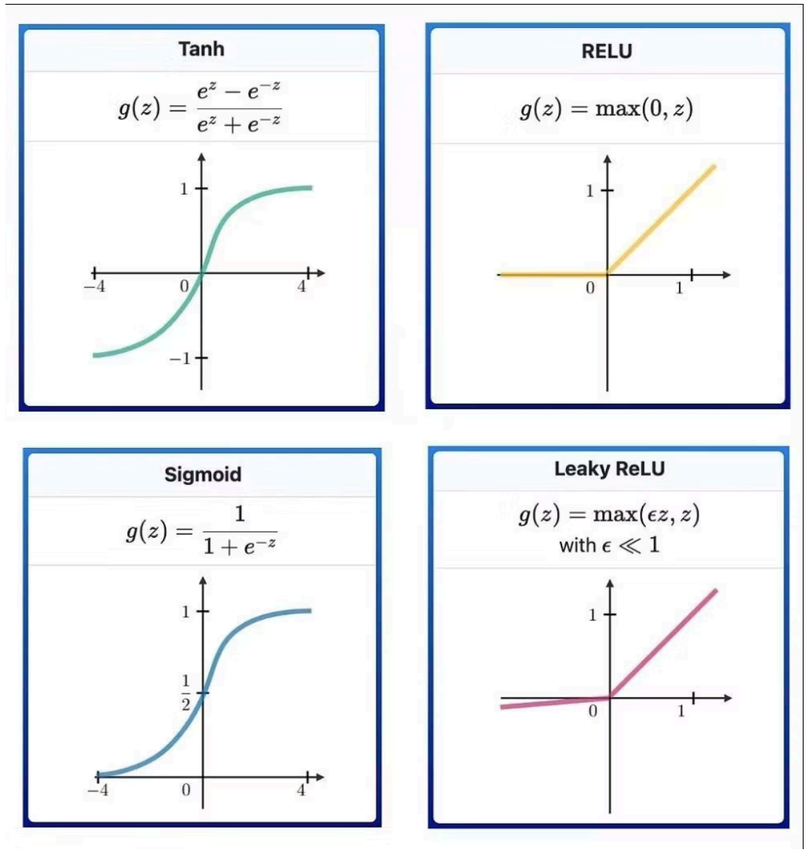
Which layer is responsible for generating the final prediction in a neural network?



Input layer	Hidden layer	Output layer	Dropout layer
0 ✗	0 ✗	8 ✓	0 ✗

8/8

What is the main role of an activation function?



0 ✓

Add non-linearity to the model

0 ×

Increase the number of neurons

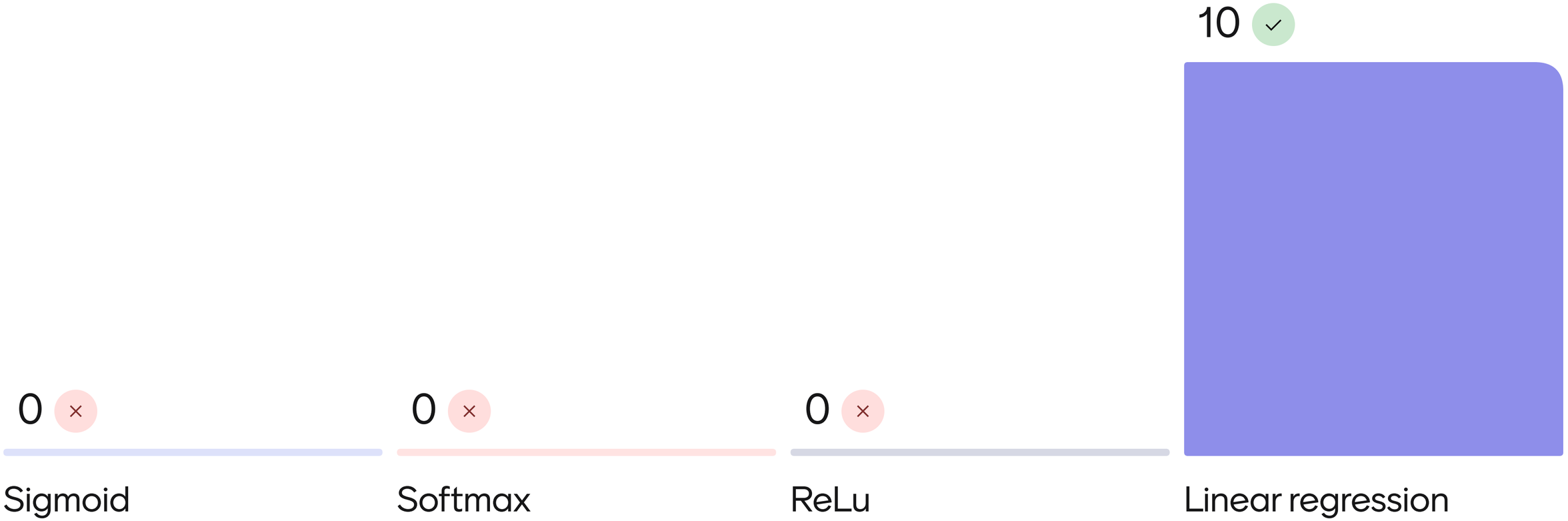
10 ×

Normalize the output

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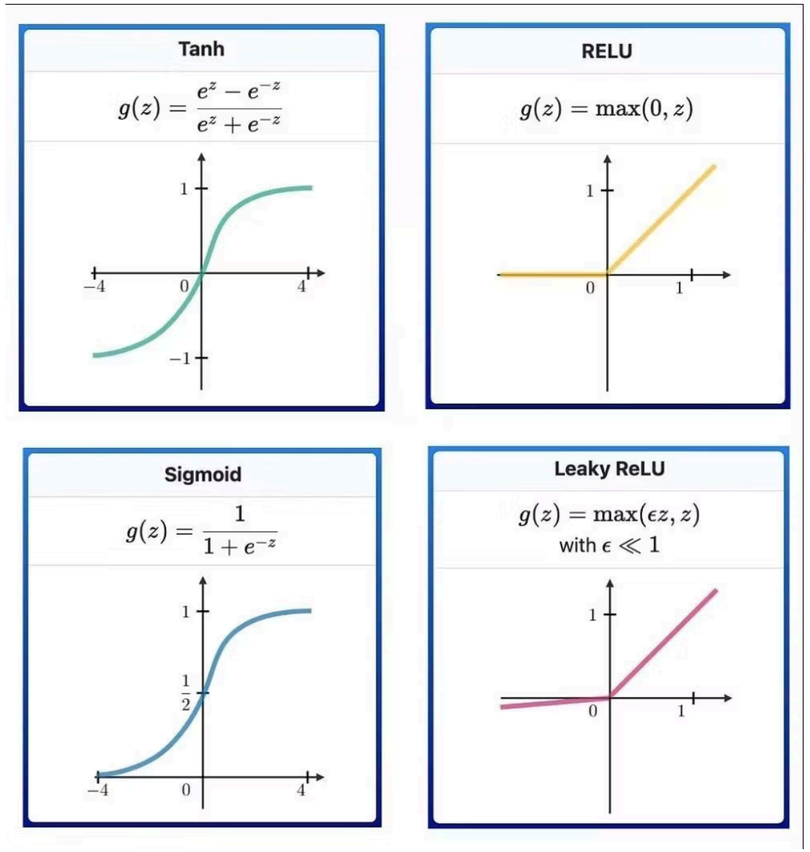
Which of the following is not a common activation function?



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Why do we use non-linear activation functions in neural networks?



9 ✓

To allow the model to learn complex patterns

0 ×

To make the network smaller

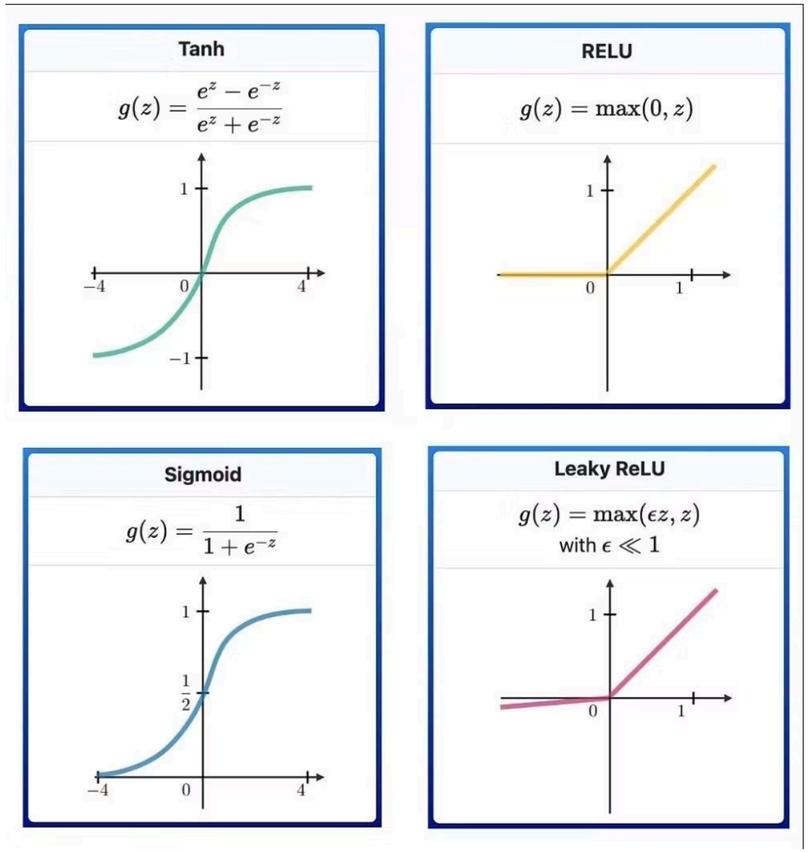
0 ×

To reduce computation

9/9

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Which activation function outputs values between 0 and 1?



7 ✓

Sigmoid

1 ×

Tanh

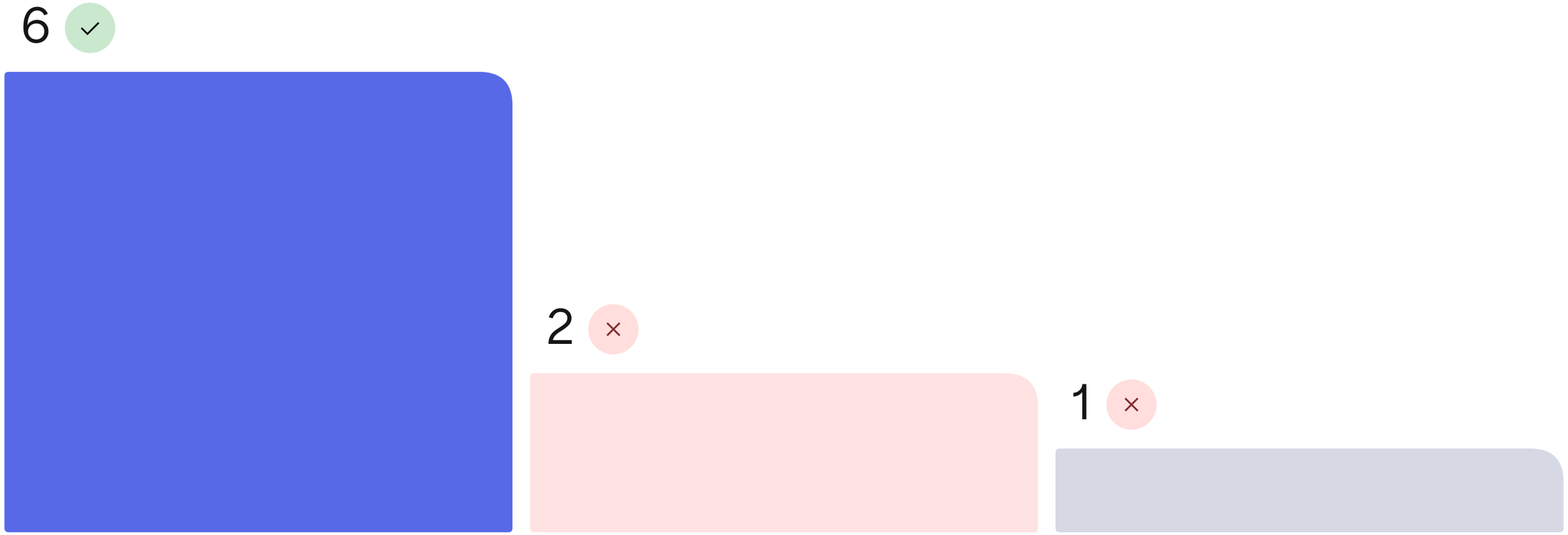
2 ×

ReLu

10/10

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What happens during forward propagation?



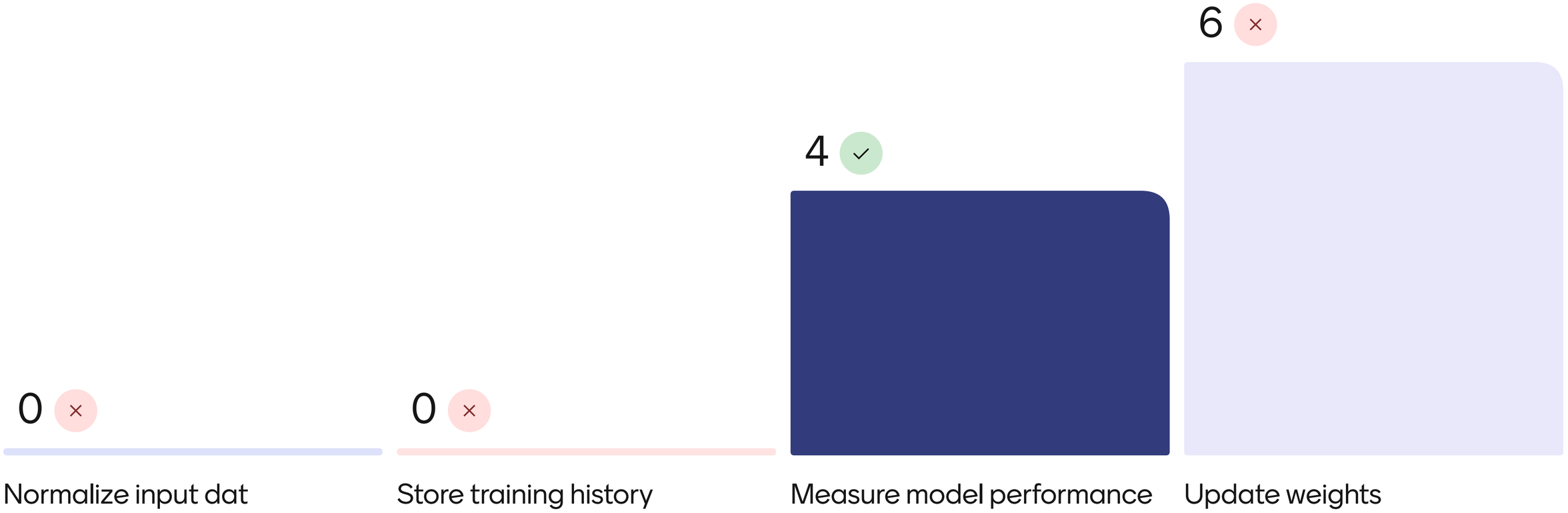
9/9

What happens during Backward propagation?



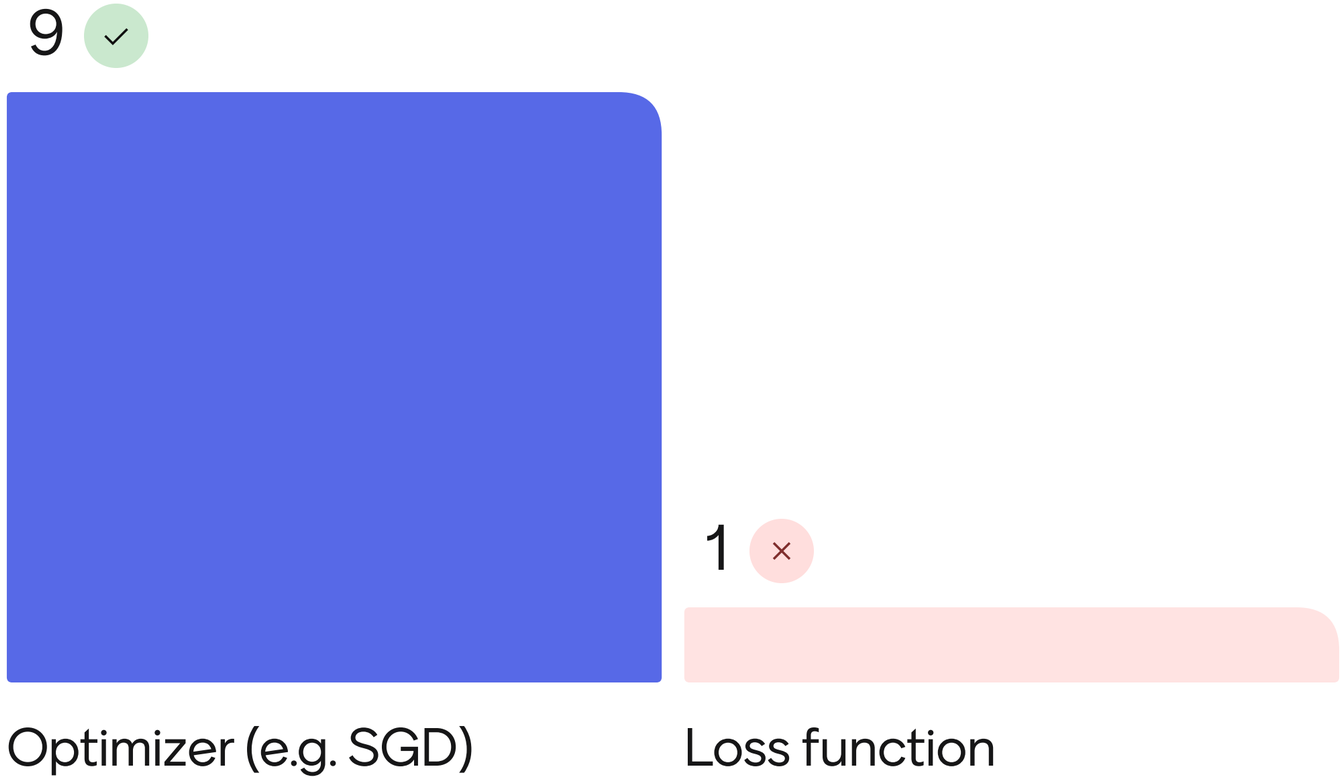
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What is the purpose of the loss function in a neural network?

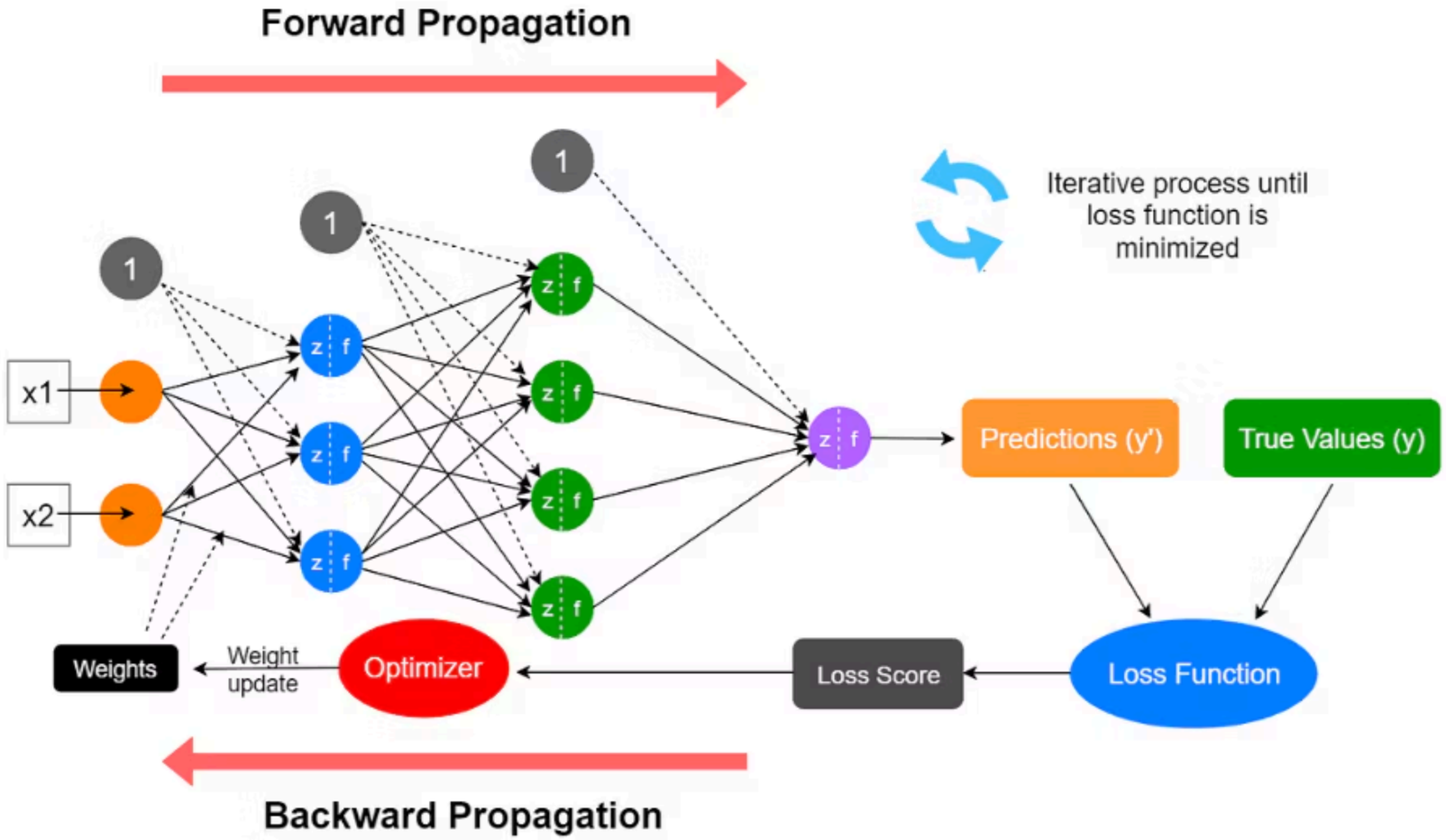


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What updates the model weights?



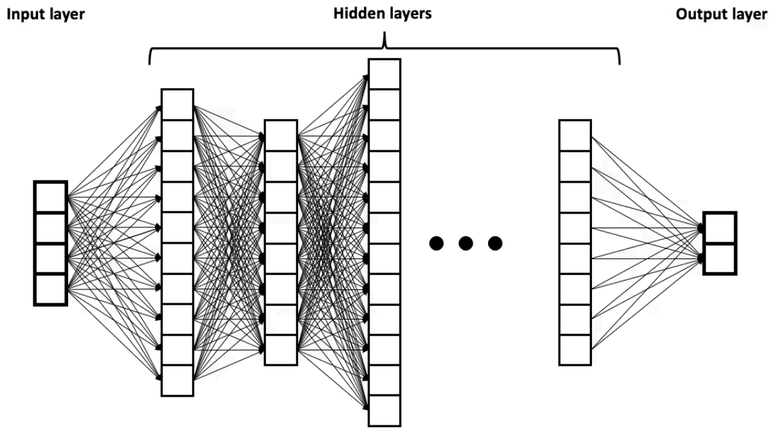
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Forward and Backward Propagation in Neural Networks

0

What happens when you increase the number of layers in a neural network?



9 ✓

The network can learn more complex patterns

0 ✗

The network becomes linear

10 ✓

It becomes more prone to overfitting if not regularized properly

0 ✗

It reduces the number of neurones

What is the purpose of dropout?

0 ×

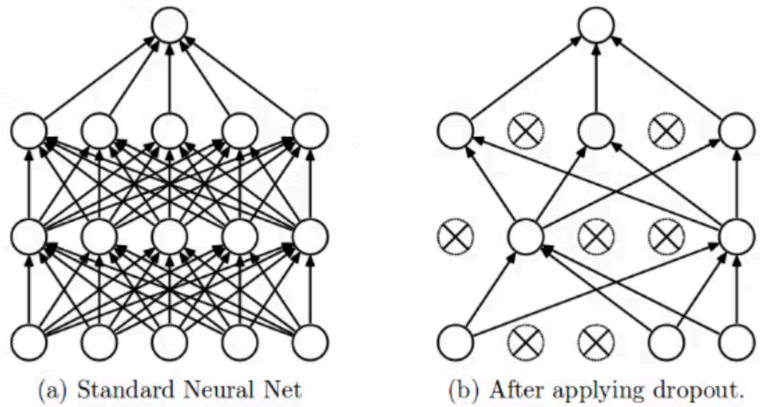
To speed up inference

1 ×

To reduce the dataset size

9 ✓

To randomly remove neurons during training to avoid overfitting



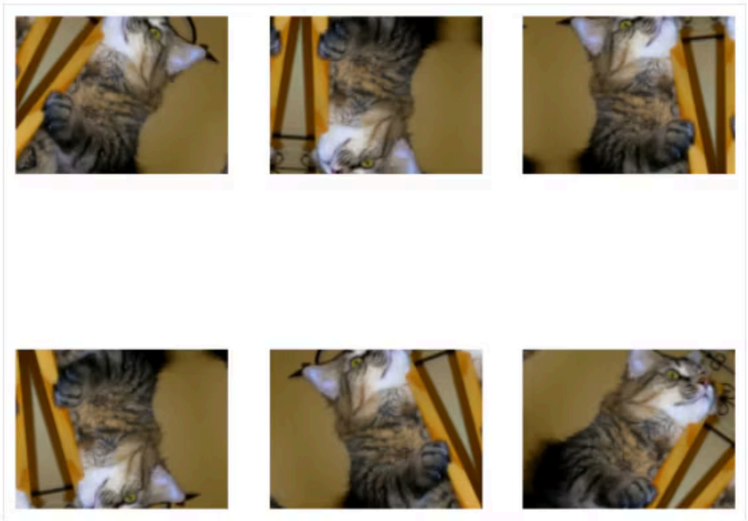
(a) Standard Neural Net

(b) After applying dropout.

10/10

Why do we use data augmentation?

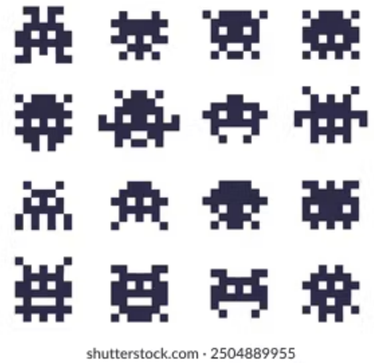


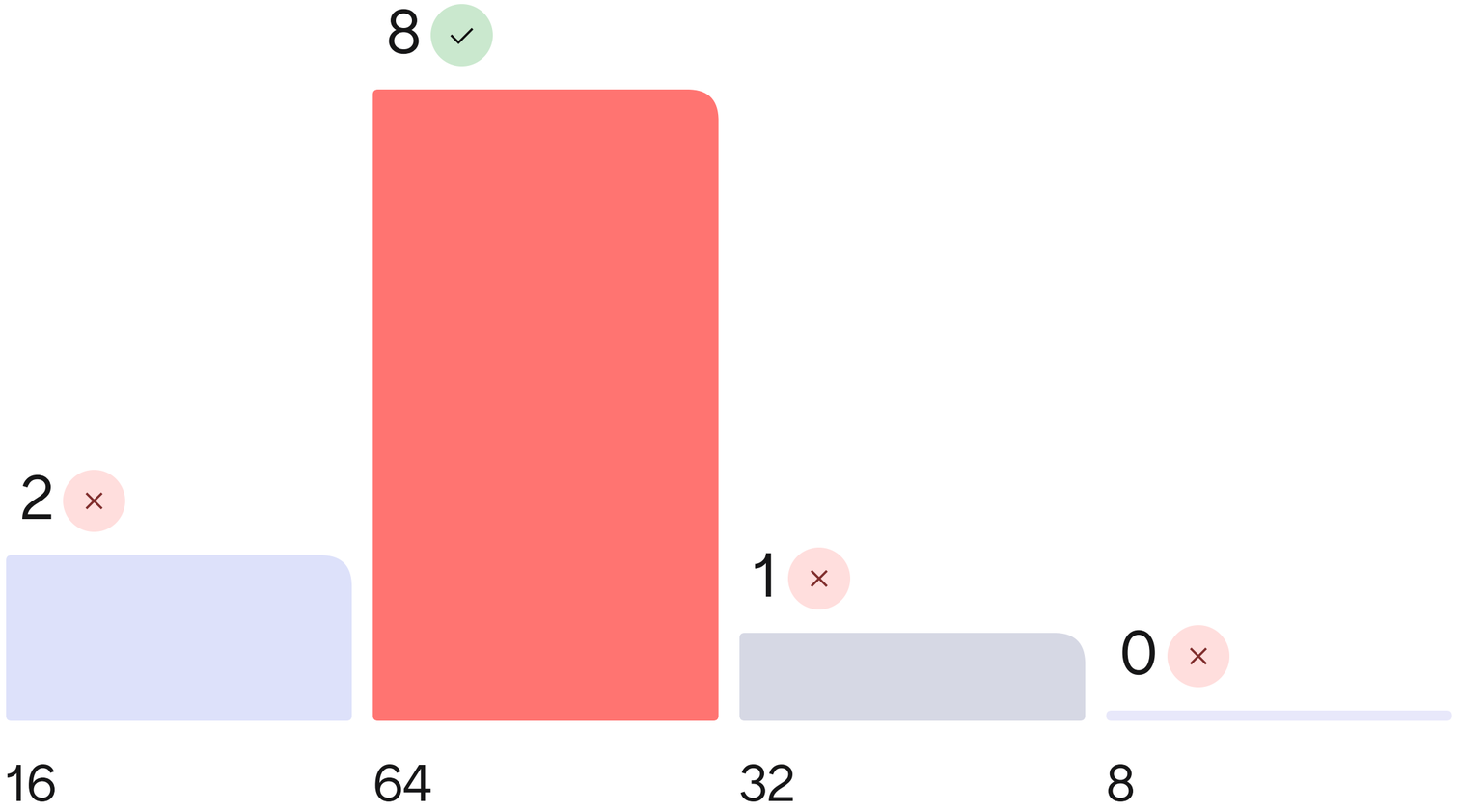


11/11

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Consider one of the following 8x8 pixel images. How many input parameters should our model have to analyze the image (the minimal reasonable amount)?





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How did you feel about the pace today?

0 questions

0 upvotes

0

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Feel free to try out the sandbox

playground.tensorflow.org

0

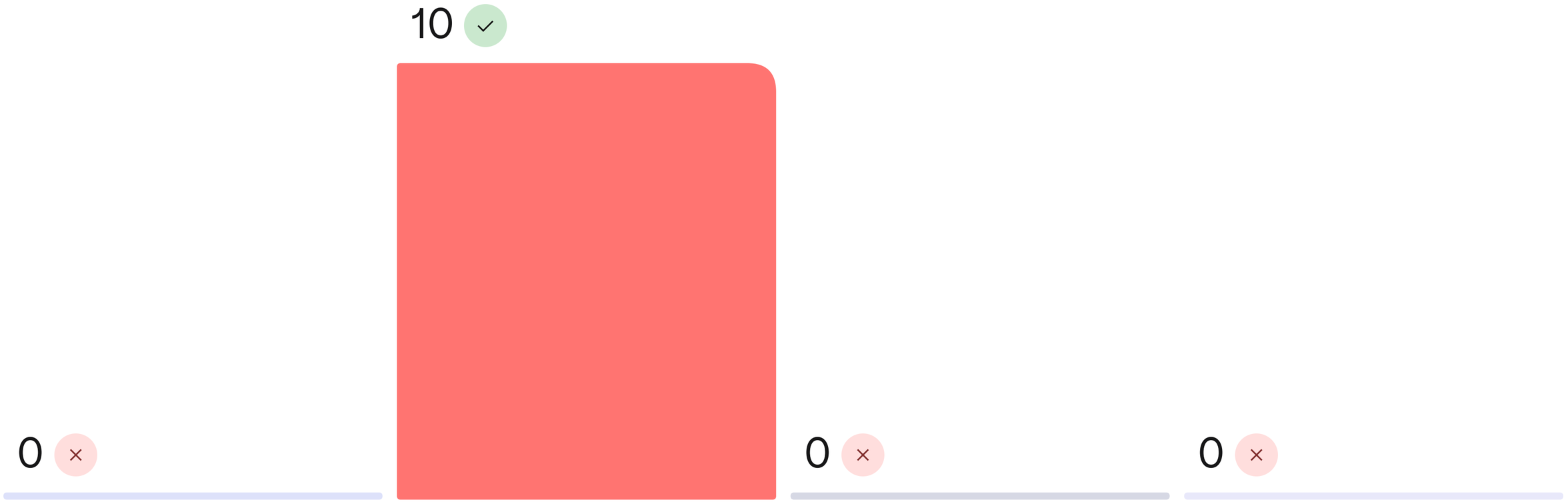
Walkthrough 4

(Text Analytics)

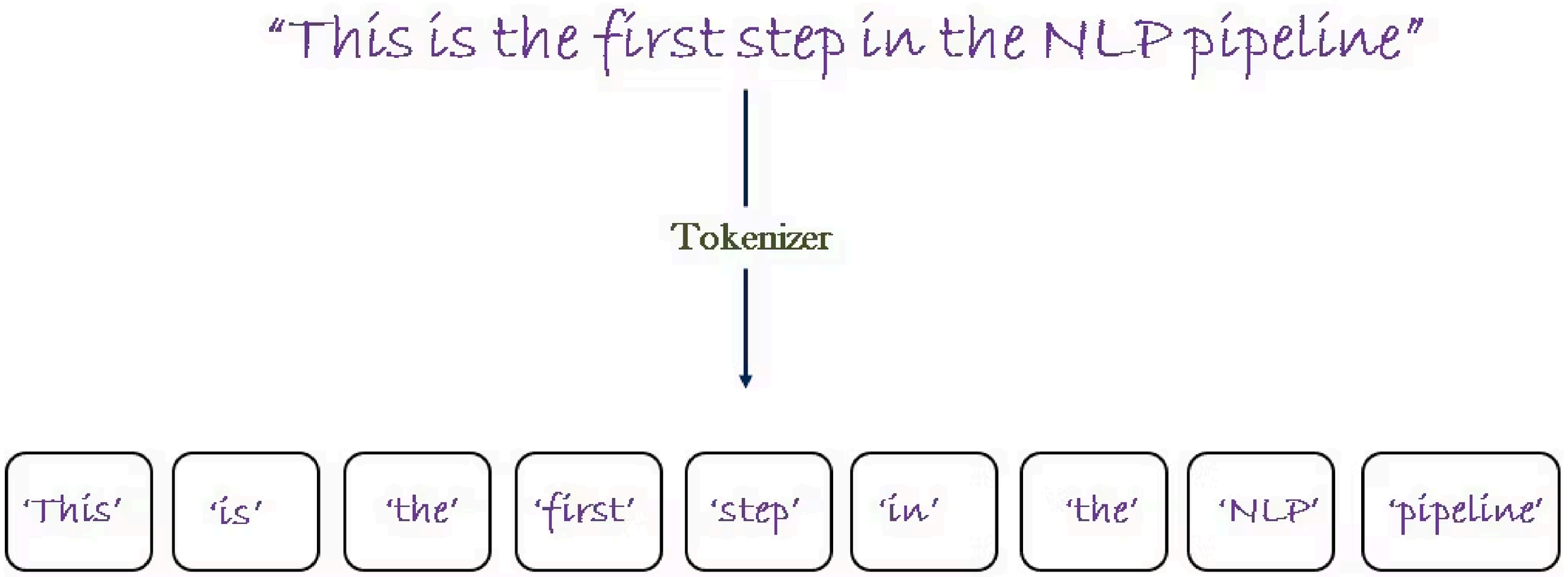
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0

What is the main goal of tokenization in text analytics?



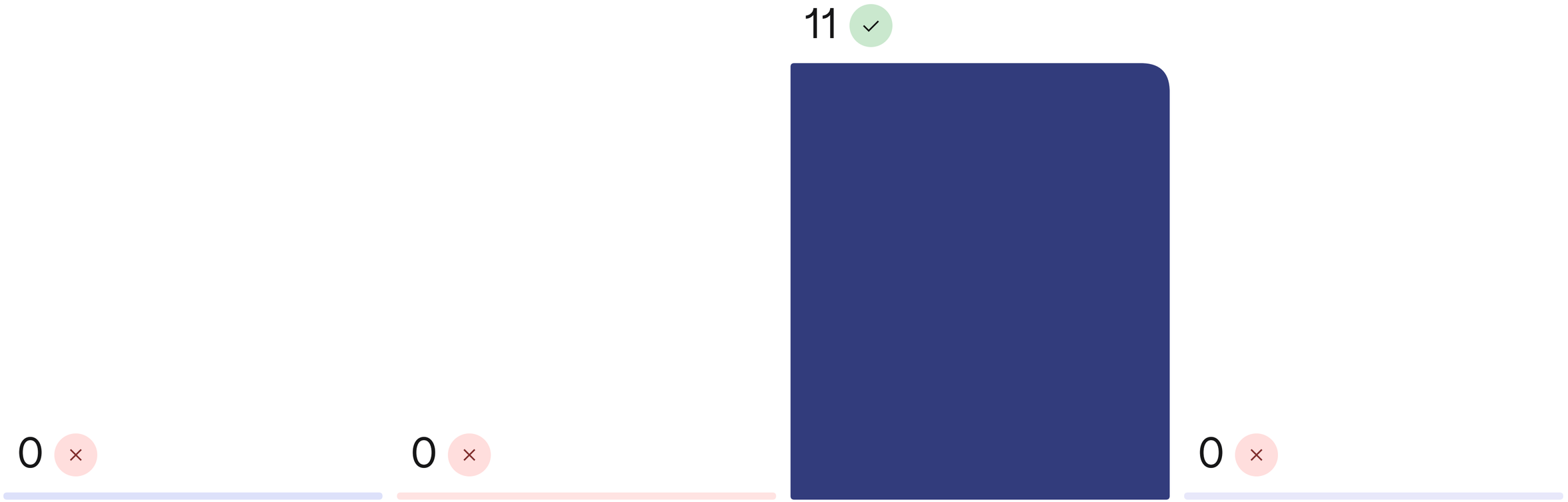
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"This is the first step in the NLP pipeline"
Tokenizer
'This'
'is'
'the'
'first'
'step'
'in'
'the'
'NLP'
'pipeline'

0

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Which of the following is a reason for removing stopwords during text preprocessing?



0 ✗

Stopwords are rare words that don't appear often in text.

0 ✗

Stopwords are words that convey essential meaning for text analysis.

11 ✓

Stopwords are common words that do not contribute much to the understanding of the text.

0 ✗

Stopwords are punctuation marks that need to be removed.

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Can you enter some stopwords ?



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Stop Words

- a
- I
- the
- in
- of
- for
- at
- to
- on
- with
- from

0

What is the key difference between lemmatization and stemming in text preparation?



8

Lemmatization reduces a word to its base form based on its meaning, while stemming simply cuts off word endings.

0

Lemmatization removes stopwords, while stemming keeps them in the text.

0

Lemmatization is a faster but less accurate method than stemming.

0

Lemmatization and stemming are two names for the same process.

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Stemming vs Lemmatization

Stemming

achieve -> achiev
achieving -> achiev

- Can reduce words to a stem that is not an existing word
- Operates on a single word without knowledge of the context
- Simpler and faster

Lemmatization

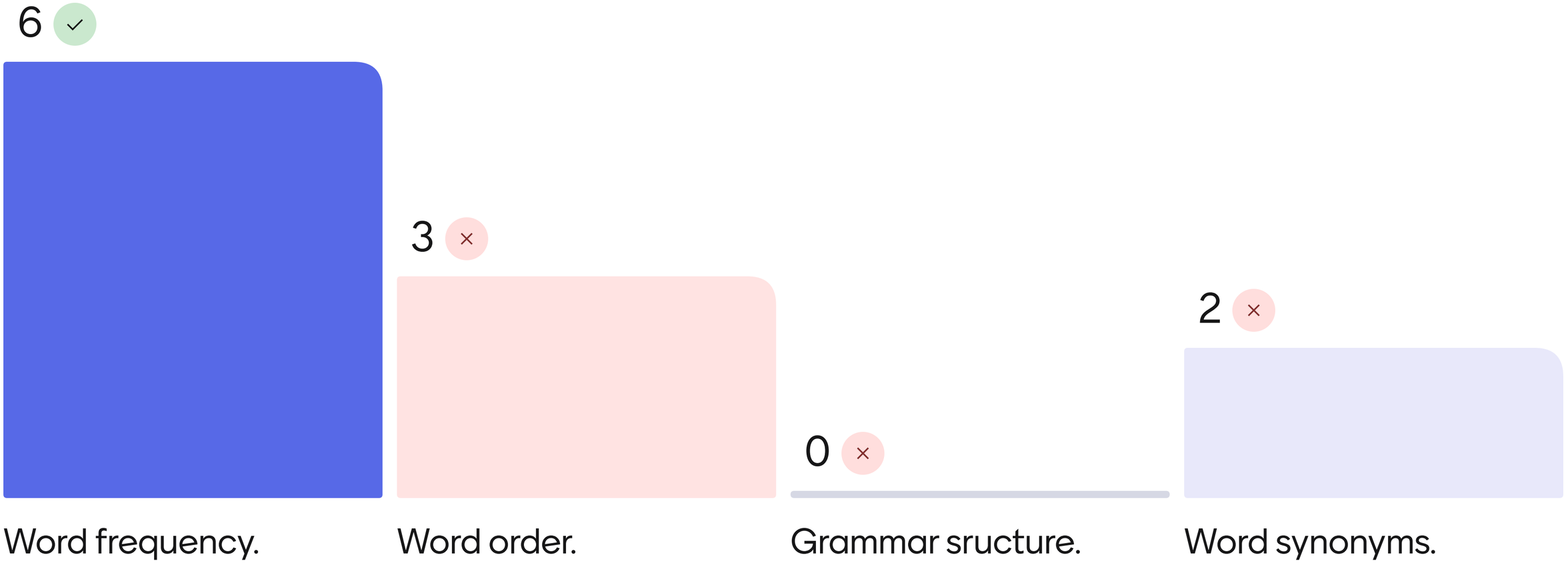
achieve -> achieve
achieving -> achieve

- Reduces inflected words to their lemma, which is always an existing word
- Can leverage context to find the correct lemma of a word
- More accurate but slower

0

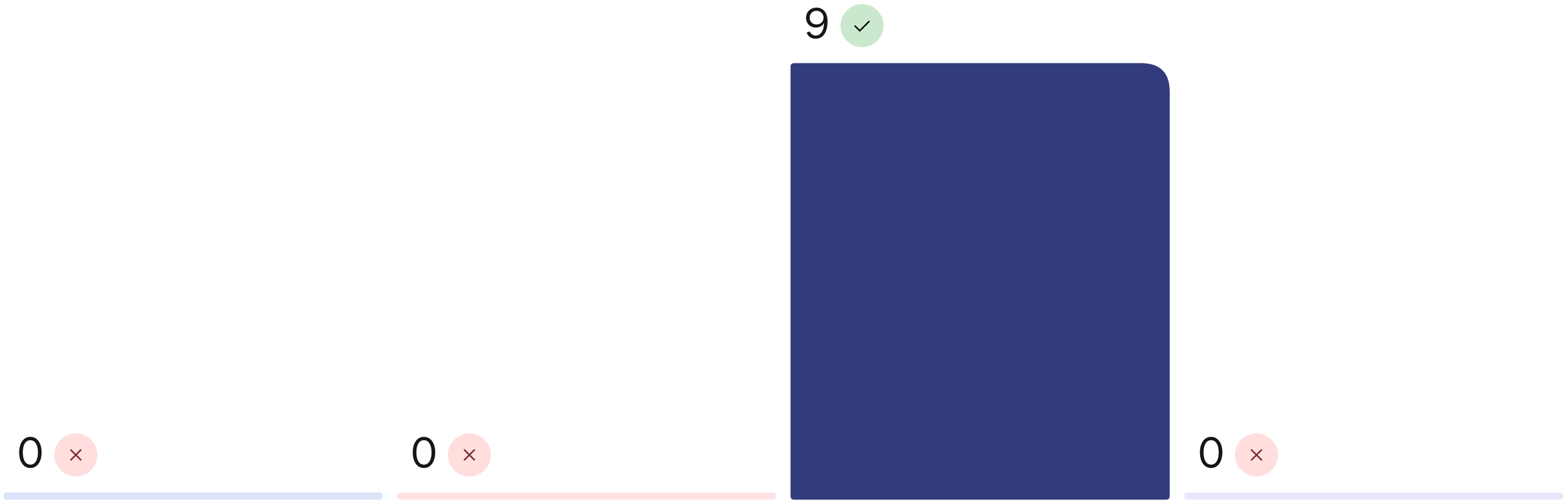
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What does Bag of Words (BOW) focus on?



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What is the purpose of the IDF (Inverse Document Frequency) component in TF-IDF?

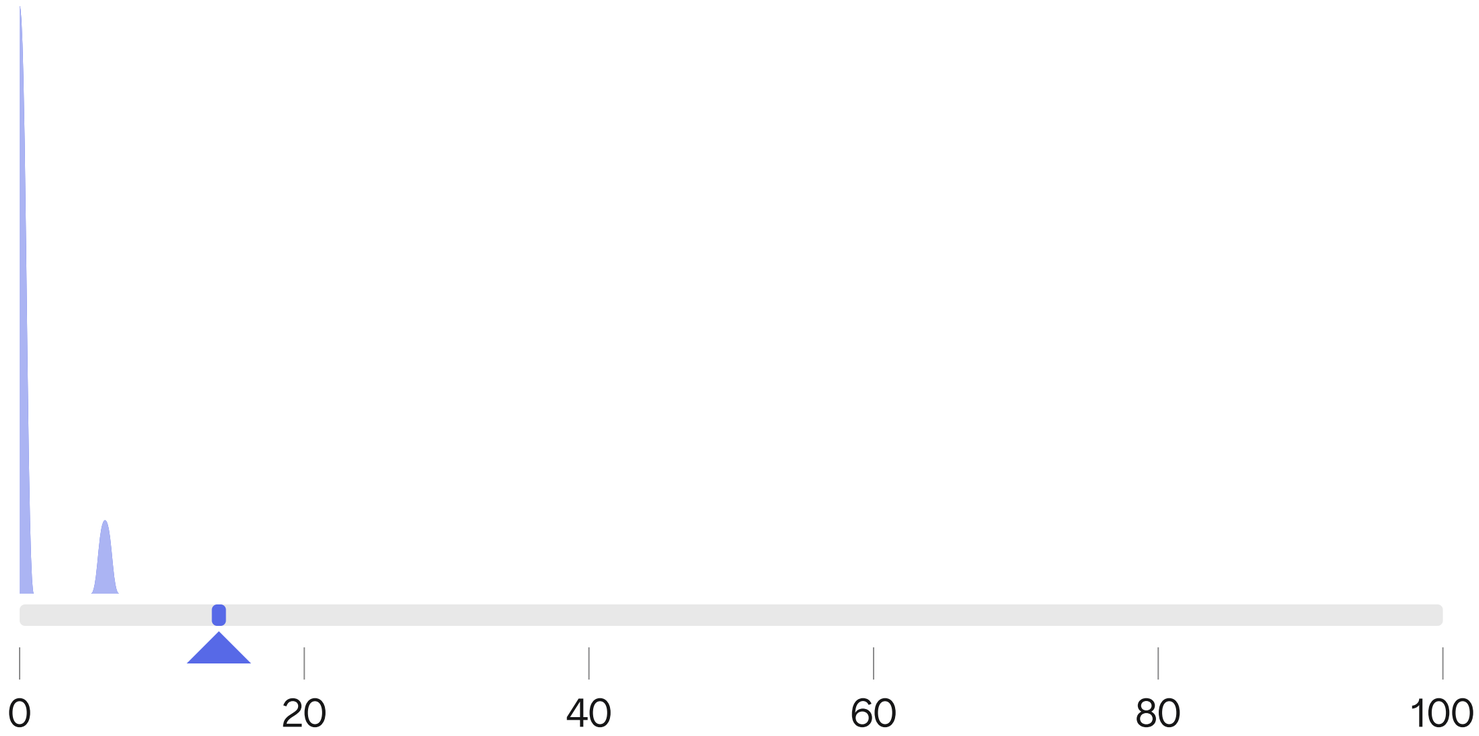


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What is the TF-IDF for the word **planet** in text A ? (e.g. 0.245 -> 25%)

14



$$w_{x,y} = tf_{x,y} \times \log\left(\frac{N}{df_x}\right)$$

TF-IDF

Term x within document y

$tf_{x,y}$ = frequency of x in y

df_x = number of documents containing x

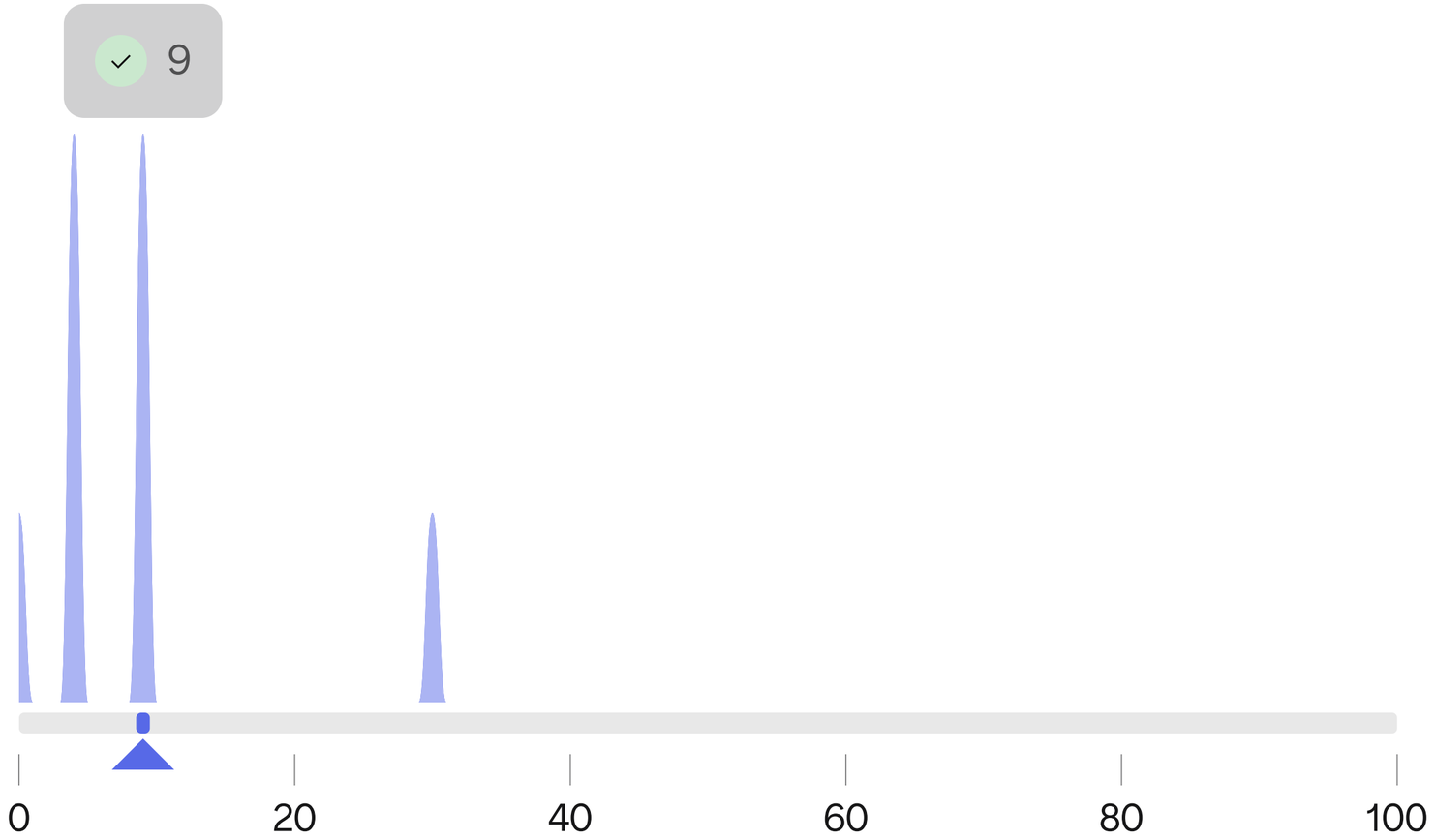
N = total number of documents

Text A: Jupiter is the largest planet

Text B: Mars is the fourth planet from the sun

9/9

What is the TF-IDF for the word **sun** in text B ? (e.g. 0.245 -> 25%)



$$w_{x,y} = tf_{x,y} \times \log\left(\frac{N}{df_x}\right)$$

TF-IDF

Term x within document y

$tf_{x,y}$ = frequency of x in y

df_x = number of documents containing x

N = total number of documents

Text A: Jupiter is the largest planet

Text B: Mars is the fourth planet from the sun

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Text A: Jupiter is the largest planet

Text B: Mars is the fourth planet from the sun

Words	TF (A)	TF (B)	IDF	TFIDF (A)	TFIDF (B)
jupiter	1/5	0	In (2/1)=0.69	0.138	0
is	1/5	1/8	In (2/2)=0	0	0
the	1/5	2/8	In (2/2)=0	0	0
largest	1/5	0	In (2/1)=0.69	0.138	0
planet	1/5	1/8	In (2/2)=0	0.138	0
mars	0	1/8	In (2/1)=0.69	0	0.086
fourth	0	1/8	In (2/1)=0.69	0	0.086
from	0	1/8	In (2/1)=0.69	0	0.086
sun	0	1/8	In (2/1)=0.69	0	0.086

0